Stock Embeddings Acquired from News Articles and Price History, and an Application to Portfolio Optimization

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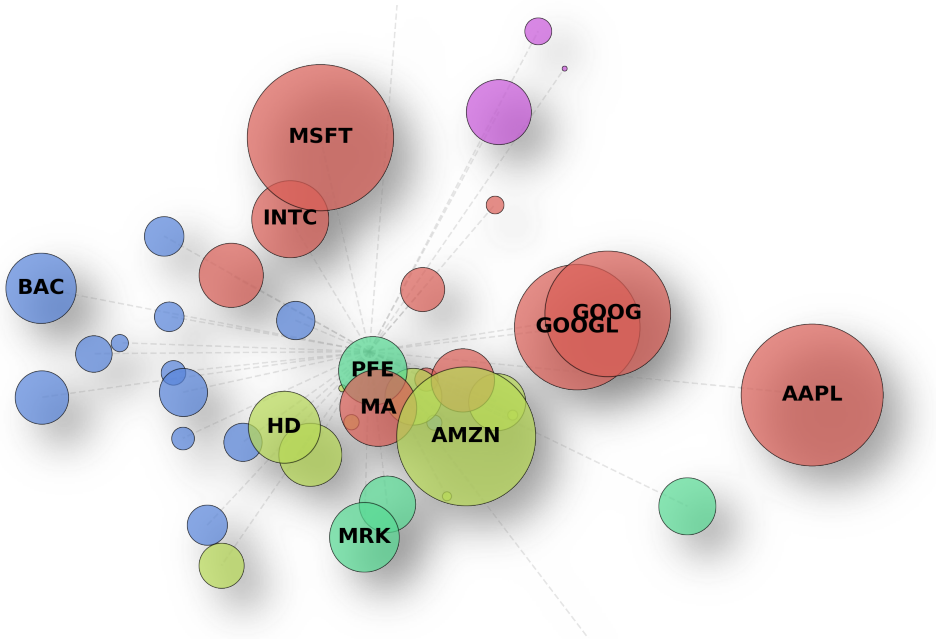

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Text vs. Price for Finance

Prices are numbers

Text provides meaningful background

- News / Financial Reports / Social media

Conventional methodology: Price → Finance

Recently watched approaches: Price + Text → Finance

Previous work

- Financial news articles. [Ou and Penman 1989] [Ding+ 2015] [Hu+ 2018] [Yang+ 2018]
- Social media texts (e.g., Twitter posts). [Nguyen+ 2015] [Xu+ 2018]

Previous Approaches

t: time j: stock

$N_{\Delta t}$: collection of news released until day t

INPUT

news articles

$N_{\Delta t}$

f

Some neural architecture

OUTPUT

price movement

$$y_t^j = \begin{cases} 1 & \text{stock } j\text{'s price up} \uparrow \\ 0 & \text{down} \downarrow \end{cases}$$

EVALUATE f (by classification accuracy)

Two problems exist

Problems

1. Market Efficiency Theory [Fama+ 1970]

   No free lunch: one can not predict more accurate than market

   ➡ Accuracy not significantly better than 50%
   (according to our re-implementation)

2. The learned result is not applicable to anything else

What we did

t: time j: stock

$N_{\Delta t}$: collection of news released until day t

INPUT

news articles

$N_{\Delta t}$

f

s_j

stock embeddings

OUTPUT

price movement

$$y_t^j = \begin{cases} 1 & \text{stock } j\text{'s price up} \uparrow \\ 0 & \text{down} \downarrow \end{cases}$$

extractable

EVALUATE f, and s_j

Stock embedding s_j is acquired via "attention"

s_j: stock embedding of stock j, learnable.

n_i: vector representation of news i, fixed.

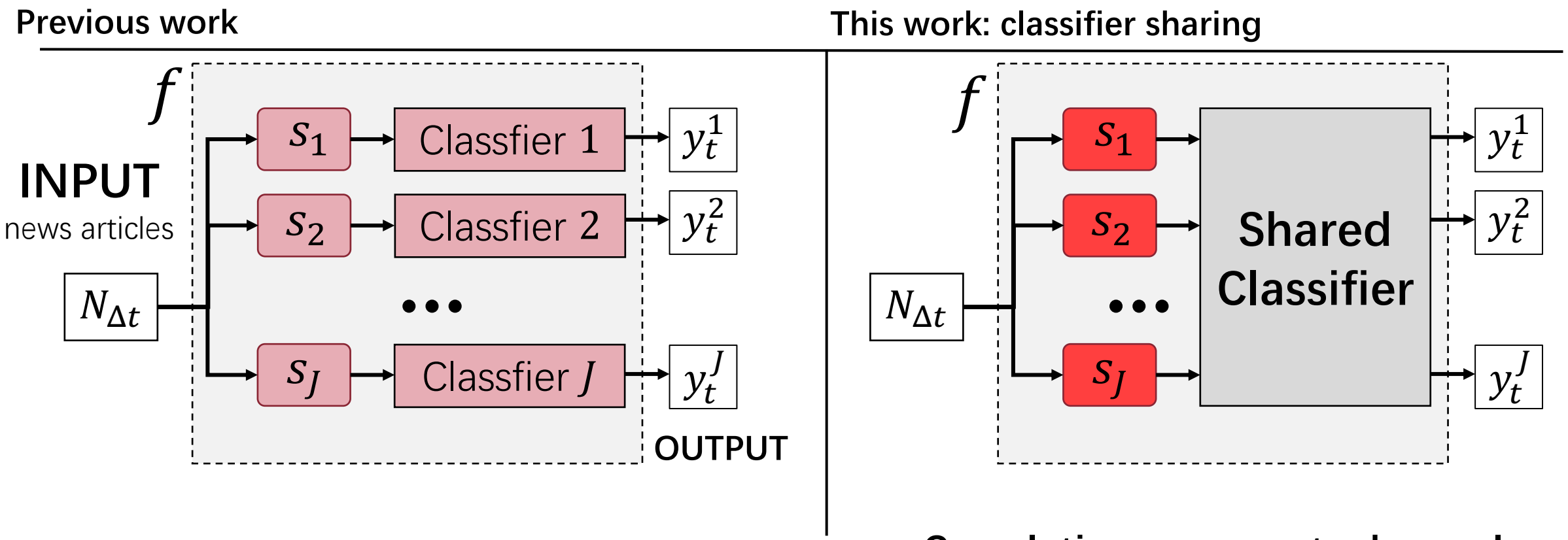
$$\text{score}_{i,j} = s_j \cdot n_i = \begin{cases} \text{high} & \text{news } i \text{ is related to } j \\ \text{low} & \text{not related} \end{cases}$$

attention

s_j finds related news n_j

Details are in the paper

Advantage: Classifier Sharing



- **Correlation among stocks are learned** and **concentrated** in s_j.
- $\boldsymbol{J}$**-times larger** training data ($J \geq 50$)
  → less overfitting

Has this solved the two problems?

Problems

1. Market Efficiency Theory [Fama+ 1970]

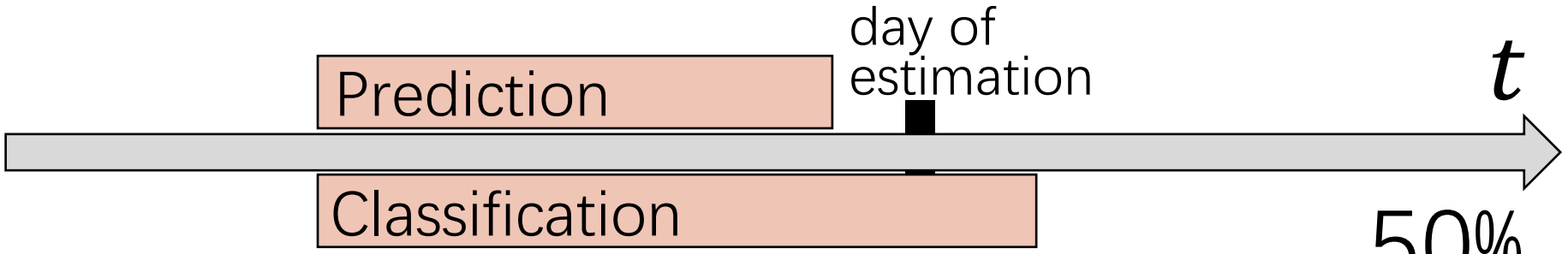
   No free lunch: one can not predict more accurate than market

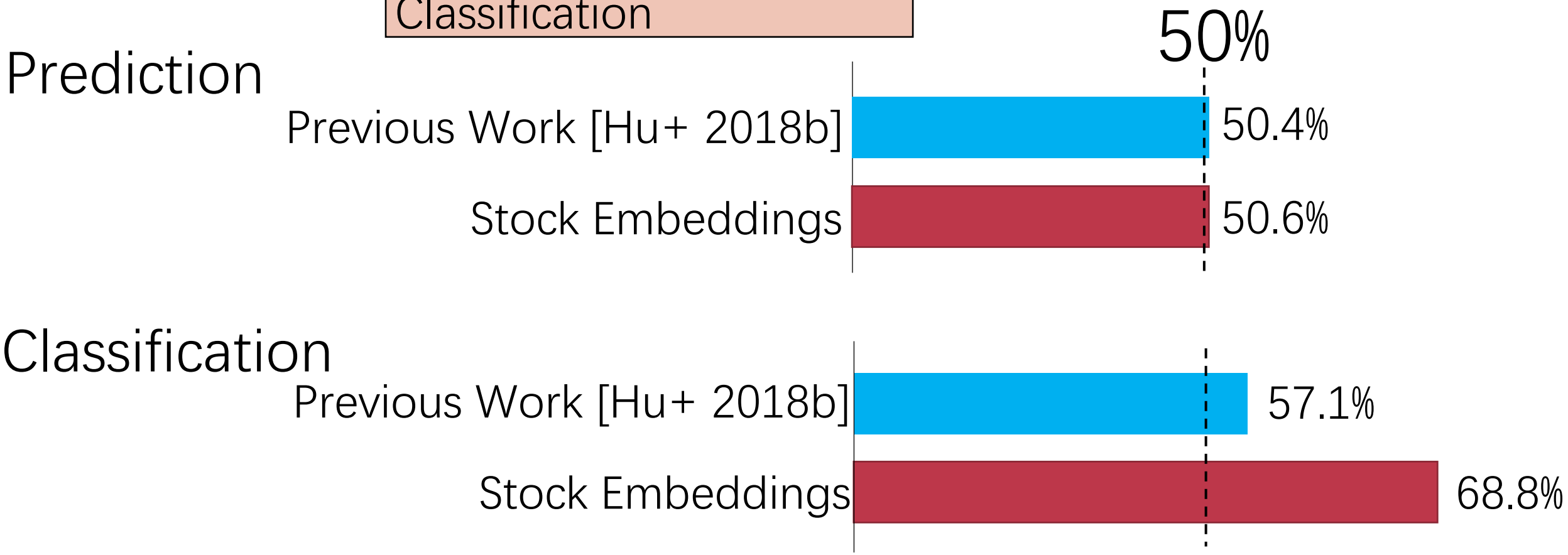
unsolved

2. The learned result is not applicable to anything else

Price prediction: Up or Down?

R & B dataset





Stock embeddings learned from texts

Problems

1. Market Efficiency Theory [Fama+ 1970]

   No free lunch: one can not predict more accurate than market

unsolved

2. The learned result is not applicable to anything else

   S_j is applicable to different problems

   → Portfolio Optimization

solved

Portfolio Optimization

Portfolio : Proportions of investment for stocks

Example
divide $ 100?

Stock	A,	B,	C,	D	
	10,	20,	30,	40	?
	90,	10,	0,	0	?

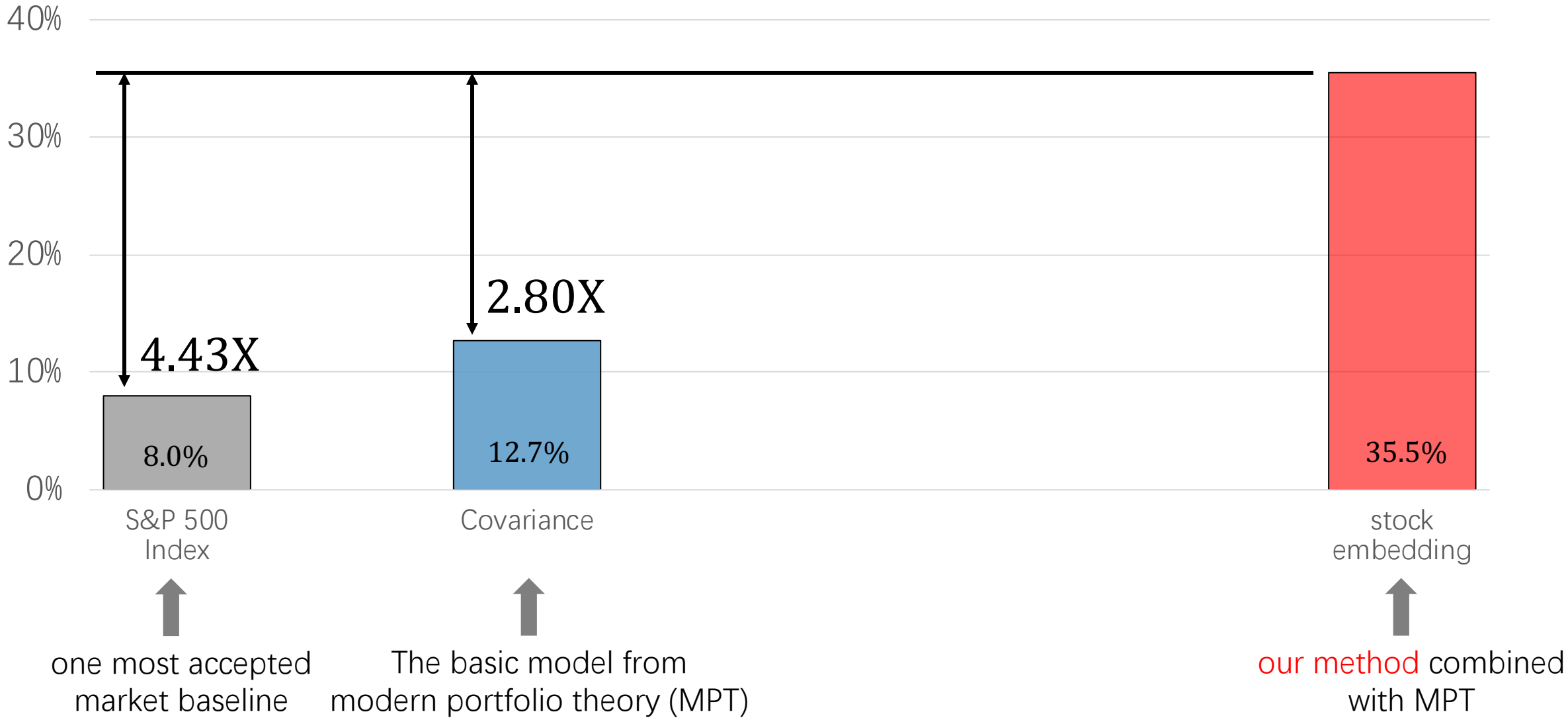
Weight vector $\boldsymbol{w}$

Markowitz's mean-variance minimization [1952]
≈ Quadratic Programming

$$\arg\min_{\boldsymbol{w}} \boxed{\boldsymbol{w}^T} \boxed{\begin{array}{c}\Sigma \\ \text{Price correlation} \\ \Downarrow \\ s_j \cdot s_k\end{array}} \boxed{\boldsymbol{w}}$$

Realized Annual Gain (R&B dataset, 2006-2013)

(averaged over years & different *E*s)



Realized Annual Gain (R&B dataset, 2006-2013)

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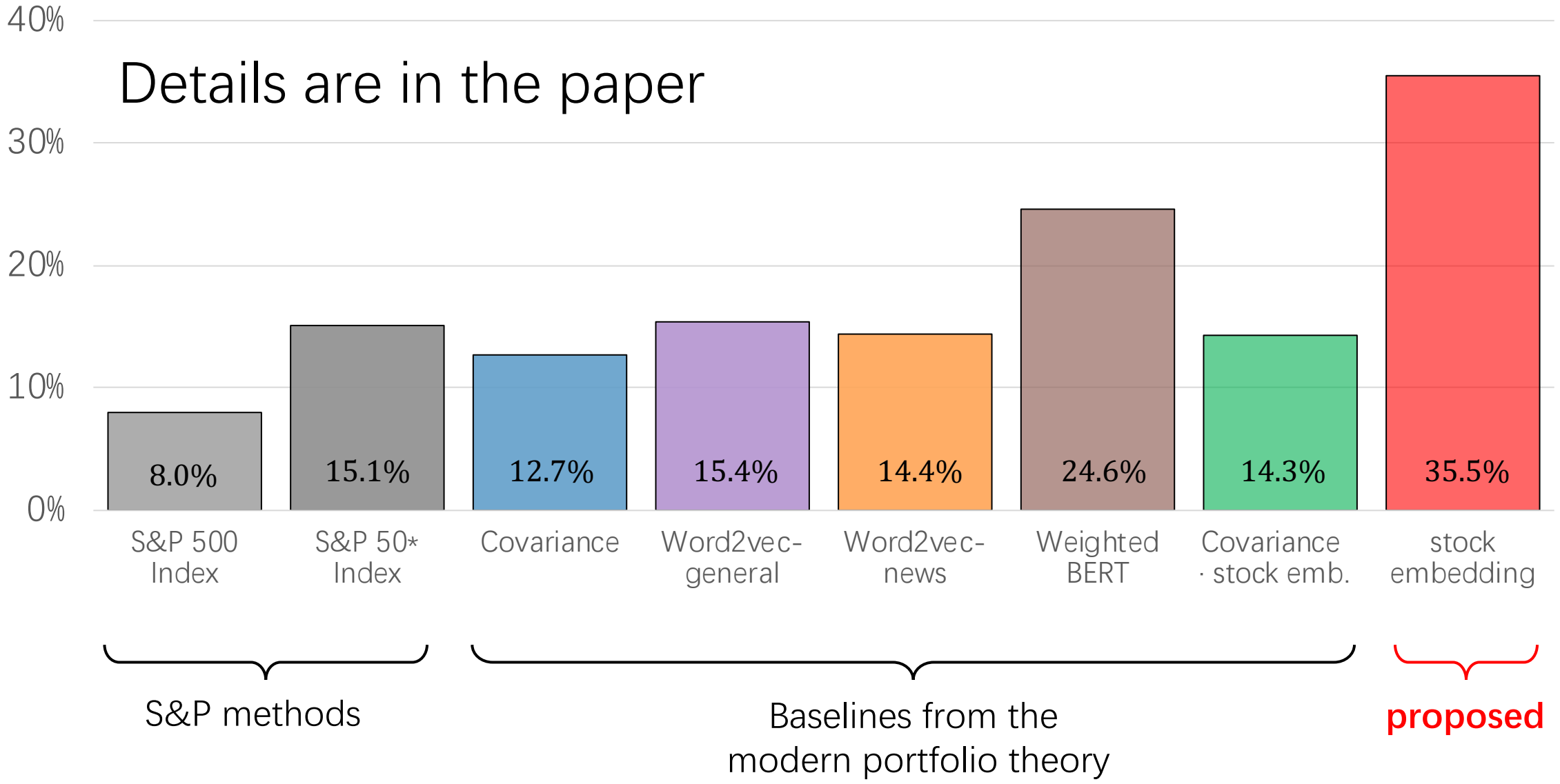
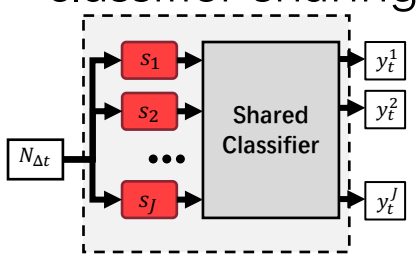


Conclusion

Stock embeddings $\vec{s}_j$ acquired from price & news text

$$\text{score}_{i,j} = \vec{s}_j \cdot \vec{n}_i = \begin{cases} \text{high} & \text{news } i \text{ is related to } j \\ \text{low} & \text{not related} \end{cases}$$

classifier sharing



Σ: correlation matrix among stocks

$\Sigma_{j,j'} = \cancel{\widehat{cov}\left(r_t^j, r_t^{j'}\right)}$ Price only

$= \cos\left(s_j, s_{j'}\right)$ Price + Text

Significant higher accuracy of price movement classification (NOT pred.)

- WSJ: $54.3\% \rightarrow 60.1\%$
- R&B $57.2\% \rightarrow 68.8\%$

Portfolio achieved up to

- 2.8X more gains than original method
- 4.4X more gains than the S&P500 index

Q&A

Time slot 1: Tuesday July 7, 2020 6B Computational Social Science and Social Media-4 6:00 UTC+0
Time slot 2: Tuesday July 7, 2020 7A Computational Social Science and Social Media-5 8:00 UTC+0

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